

Mahima Kaur

mahimakaur20@gmail.com | LinkedIn | GitHub | Google Scholar

Data Engineer with 3+ years of experience building data pipelines and analytics solutions using Python and SQL. Skilled in end-to-end ELT pipelines, data modeling, and workflow orchestration for large-scale datasets. Familiar with AWS and expanding cloud expertise. Open to relocation to Denver, Colorado.

Education

Yale University Master of Science, Health Informatics	New Haven, CT, USA May 2024
University of Delhi Master of Science	New Delhi, India May 2019

Experience

Harvard T.H. Chan School of Public Health Data Engineer	July 2024 – present
- Owned end-to-end ELT pipelines in Python and SQL to process 10+ TB of multi-year data within a data lakehouse architecture, from ingestion to analytics-ready datasets. - Defined data models and standardized variables to ensure consistency and usability of analytical datasets across teams. - Built and maintained scalable data workflows using Snakemake and Hydra, running on HPC clusters (SLURM) and optimized with DuckDB over partitioned Parquet. - Developed deployable data products using containerized workflows (Docker) for data processing pipelines and dashboards. - Enhanced data performance and accessibility by implementing partitioning strategies and establishing a centralized metadata-driven data catalog.	
Yale University-Library StatLab StatLab Consultant	May 2023 – May 2024
Provided statistical consulting to students and researchers on data analysis, coding, and visualization using Python, R, and SAS.	
Foundation of Healthcare Technologies Society Research Associate	July 2019 – April 2022
- Conducted statistical analyses and hypothesis testing to generate and validate research findings. - Developed data visualizations and reports using R to communicate insights to non-technical stakeholders. - Designed and maintained data processing workflows for data ingestion, cleaning, and validation, ensuring reliable and consistent datasets across studies.	

Projects

Diabetes Risk Calculator
Built a diabetes risk prediction model on 70K records and deployed a dashboard using Flask and Plotly.

Conference Talk

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- Kaur, M.,** Nalluri, S., C. Kitch, J., M. Tapera, T., Gilmour, J., Audirac, M., & Braun, D. (2025, October 6). A reproducible and scalable pipeline for processing administrative health claims data. USRSE'25. Zenodo. <https://doi.org/10.5281/zenodo.17281686>

Skills

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- Languages:** Python, SQL, R, Bash
 - Pipelines & Orchestration:** Airflow, Dagster, Snakemake, Hydra, dbt, ETL/ELT
 - Data Modeling:** star schema, dimensional modeling, data lakehouse, materialized views
 - Warehousing & Storage:** Redshift, Snowflake, BigQuery, PostgreSQL, DuckDB, Parquet
 - Infrastructure:** Docker, Git, CI/CD, Linux, HPC, SLURM
 - Visualization:** Looker, Tableau, Power BI